

Satellite Structural Inspection Interferometric Synthetic Aperture Radar (InSAR) Based

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METHODOLOGY

1. Satellite SAR System with Resolution and Revisit Time

To select a suitable SAR system, technical specifications such as spatial resolution, frequency band, polarization, and revisit time were considered. Sentinel-1, operating at C-band (5.405 GHz), was selected due to:

Parameter	Sentinel-1 C-SAR (IW Mode)
Frequency Band	C-band (~5.405 GHz)
Revisit Time	12 days (single satellite)
Spatial Resolution	5 m (range) × 20 m (azimuth)
Coverage Mode	Interferometric Wide Swath (IW)

2. Retrieving Multi-Temporal SAR Imagery Over the Specified Period

Multi-temporal **Sentinel-1 SLC IW mode** data were downloaded based on:

- **Area of Interest (AOI):** A polygon covering the bridge and surrounding terrain.
- **Date range:** Multiple acquisitions were collected spanning several months to capture temporal variations.
- **Image acquisition conditions:** All images were selected with the same relative orbit, pass direction (ascending or descending), and polarization (typically VV/VH) to ensure consistency for interferometric analysis.

S1A_IW_SLC__1SDV_20250111T130240_20250111T130310_057397_0710AB_813D
S1A_IW_SLC__1SDV_20250123T130239_20250123T130309_057572_07179E_5431
S1A_IW_SLC__1SDV_20250204T130238_20250204T130308_057747_071E86_3FE6
S1A_IW_SLC__1SDV_20250216T130238_20250216T130308_057922_072592_C312
S1A_IW_SLC__1SDV_20250228T130238_20250228T130307_058097_072CB4_7A14
S1A_IW_SLC__1SDV_20250312T130238_20250312T130308_058272_0733CA_217F
S1A_IW_SLC__1SDV_20250324T130238_20250324T130308_058447_073AB5_B445
S1A_IW_SLC__1SDV_20250405T130238_20250405T130308_058622_0741CE_2B52
S1A_IW_SLC__1SDV_20250417T130238_20250417T130308_058797_0748F4_3FFD
S1A_IW_SLC__1SDV_20250429T130239_20250429T130309_058972_07501E_08D8
S1A_IW_SLC__1SDV_20250511T130238_20250511T130308_059147_0756BA_DE13
S1A_IW_SLC__1SDV_20250523T130238_20250523T130308_059322_075CCB_14B2
S1A_IW_SLC__1SDV_20250604T130237_20250604T130307_059497_0762D9_3AB5
S1A_IW_SLC__1SDV_20250616T130237_20250616T130306_059672_0768D1_F57D
S1A_IW_SLC__1SDV_20250628T130236_20250628T130306_059847_076EE3_6BFD

3. Differential SAR Interferometry (D-InSAR) for Displacement Measurement

D-InSAR analysis was performed to measure millimetre-scale deformation over time:



VISUALIZATIONS

1. Defining specific bridge location & parameters for satellite SAR based inspection



Project & Bridge Details

State – Rajasthan

Bridge No. – 899/1

Bridge Chainage – 914+260

Project - Chittorgarh to Kota

Zone – RJ 06

Structure Type – MNB

Sar Parameters

Sensor – Sentinel-1 (C-Band)

Beam Mode – IW (Interferometric Wide Swath)

Path – 34

Frame – 519

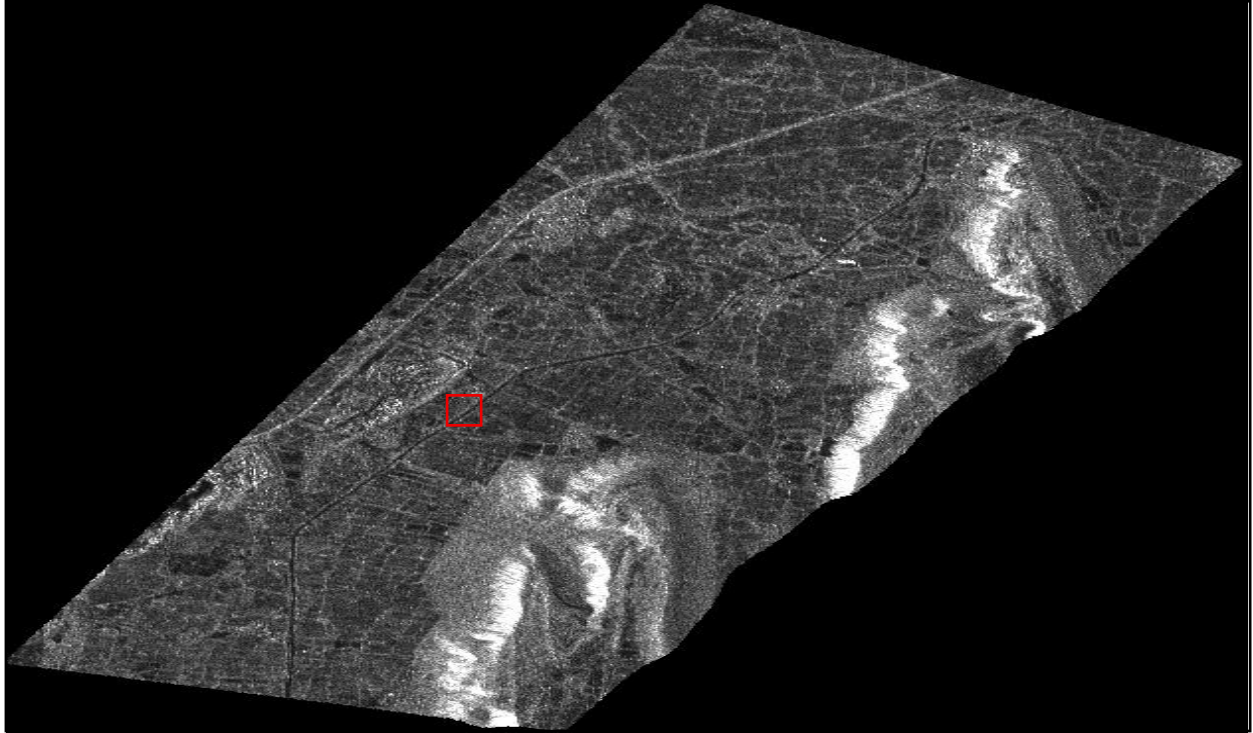
Flight Direction - DESCENDING

Polarization – VV+VH

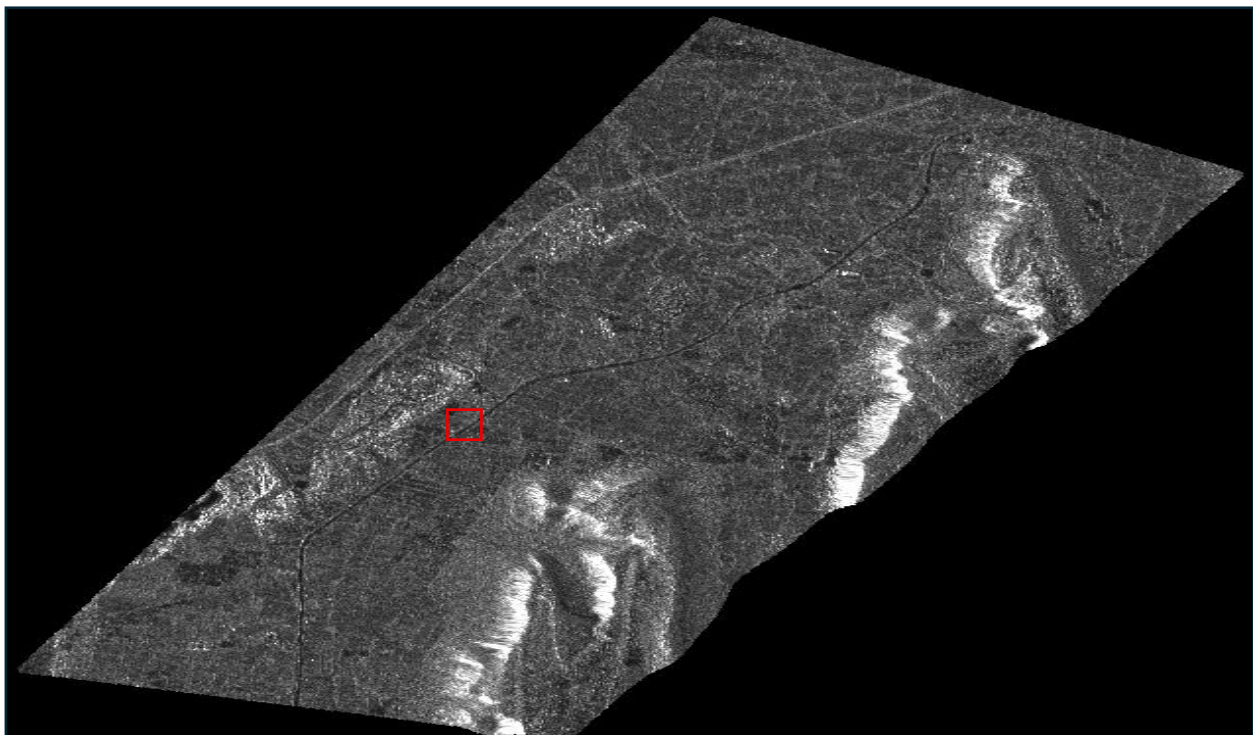
Absolute Orbit – 54706

2. Calibration, georeferencing, orthorectification of SAR images (VH/VV)

- Orthorectification of SAR image VH Polarization



- Orthorectification of SAR image VV Polarization



Bridge Components

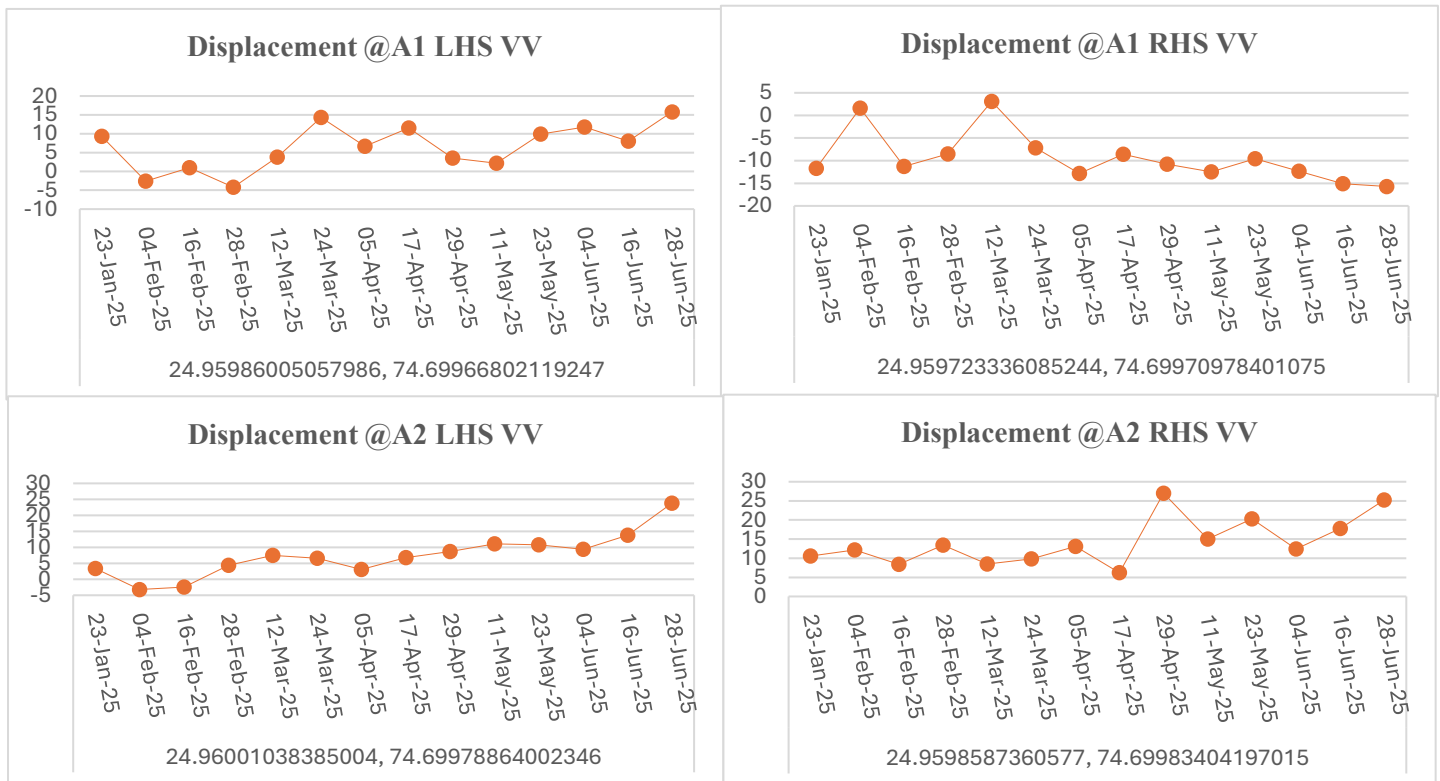
MNB – CH-914+260

Length – 20.2M

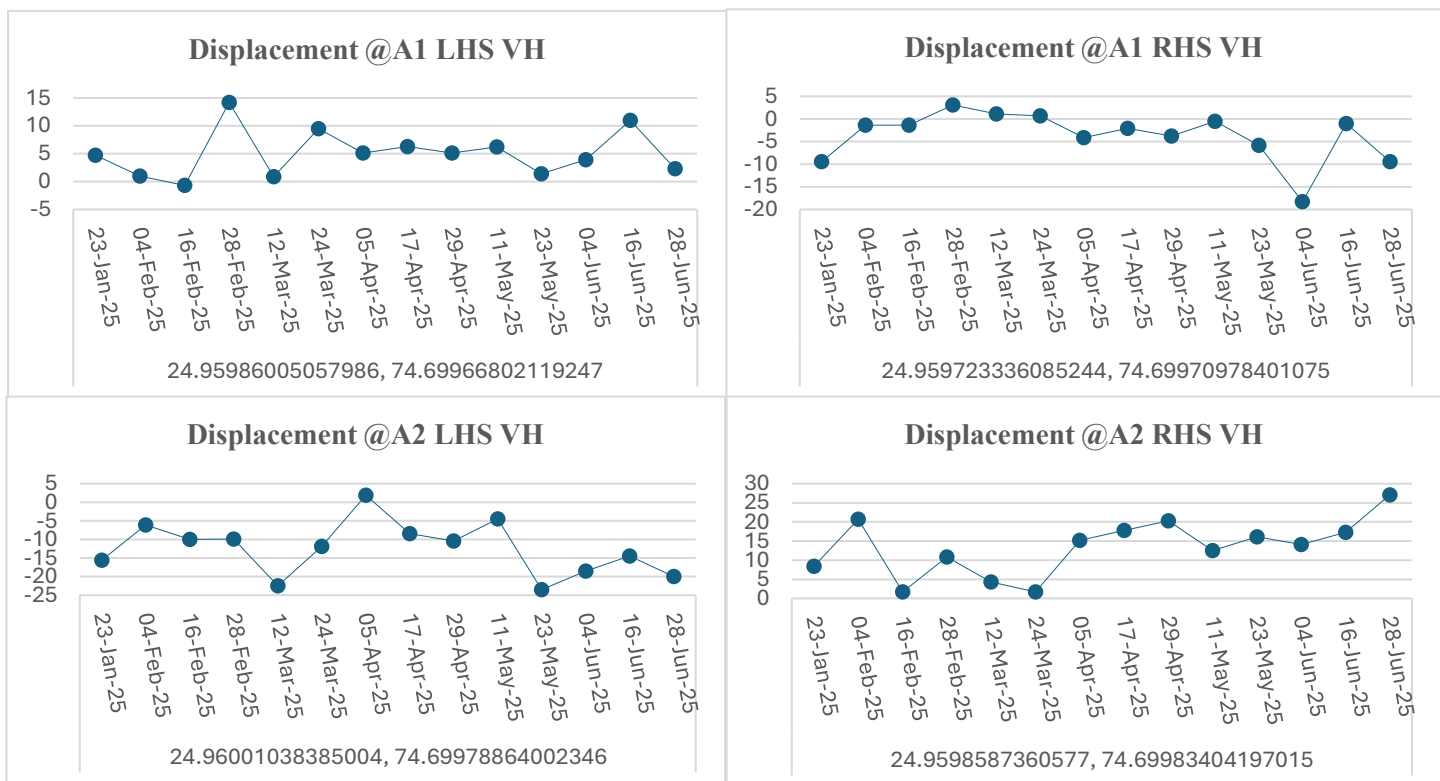


Scan Duration		From : 23 Jan 2025	To : 28 June 2025
Components & Abbreviation			
A1	Abutment 1		
A2	Abutment 2		
LHS	Left Hand Side		
RHS	Right Hand Side		

3.1 VV Displacement From 23 Jan 2025 To 28 June 2025



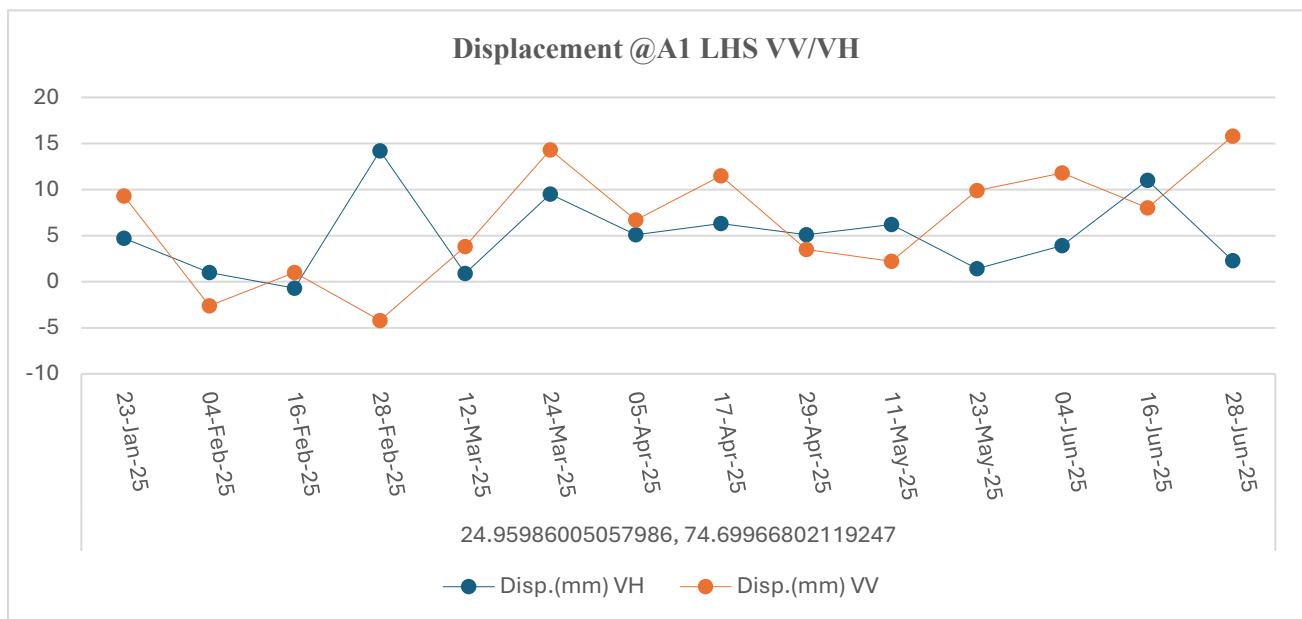
3.2 VH Displacement From 23 Jan 2025 To 28 June 2025



3.3 SAR Based Analysis at Abutment 1 LHS:



A1 LHS (24.95986005057986, 74.69966802119247)



Key Observations:

Displacement Magnitude Patterns:

- **Vertical Displacement (VV)** Vertical displacement exhibits moderate variability with alternating uplift and settlement phases over the monitoring period (Jan–Jun 2025). VV values range approximately from –4 mm to +15 mm.
- **Horizontal Displacement (VH)** Horizontal displacement shows moderate variability with both positive and negative excursions, ranging roughly between –1 mm and +14 mm.

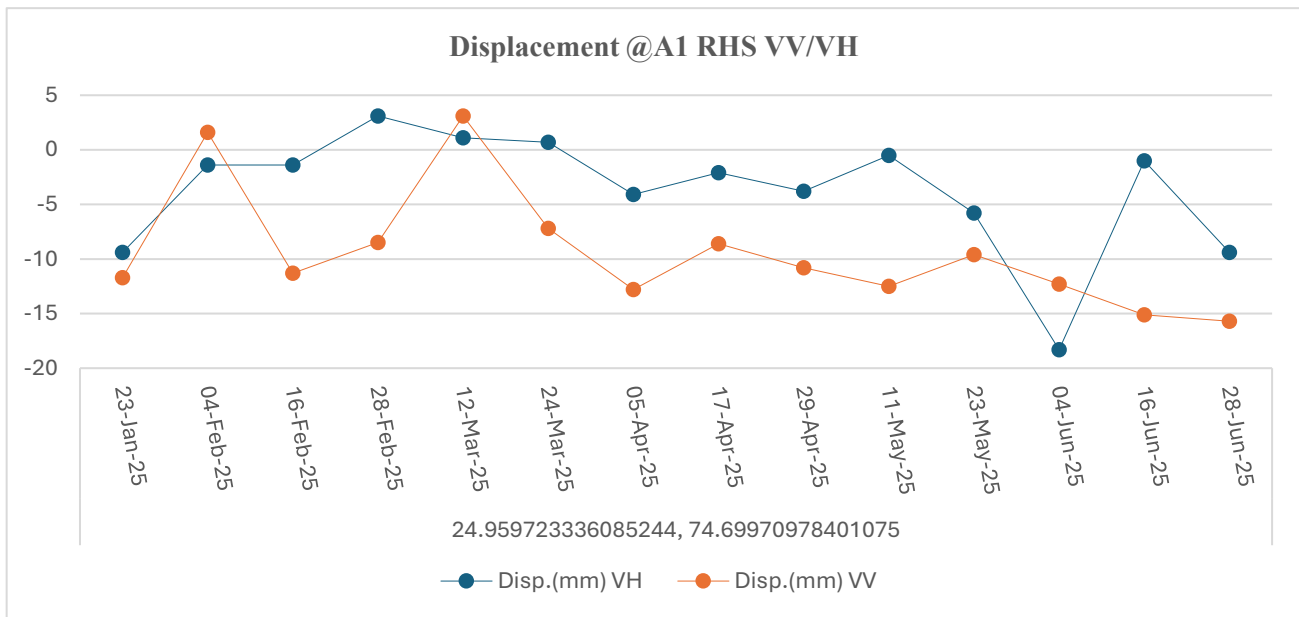
Critical Events & Anomalies:

- **March 2025:** VV peaks near +14 mm, while VH increases to +9 mm, indicating a strong coupled uplift event.
- **April 2025:** VV drops briefly to +3 mm, while VH remains positive (+6 mm), reflecting a minor vertical adjustment under continued lateral displacement.
- **June 2025:** VV increases steadily to +15 mm, while VH fluctuates between +3 and +11 mm, confirming continued uplift-dominated vertical behaviour with intermittent lateral variability toward the end of the monitoring period.

3.4 SAR Based Analysis at Abutment 1 RHS:



A1 RHS (24.959723336085244, 74.69970978401075)



Key Observations:

Displacement Magnitude Patterns:

- **Horizontal displacement (VH)** Horizontal displacement exhibits moderate to high variability with alternating uplift and settlement phases, ranging roughly between +3 mm and –18 mm.
- **Vertical displacement (VV)** Vertical displacement remains predominantly negative throughout the monitoring period, indicating overall vertical settlement at the A1 RHS location.

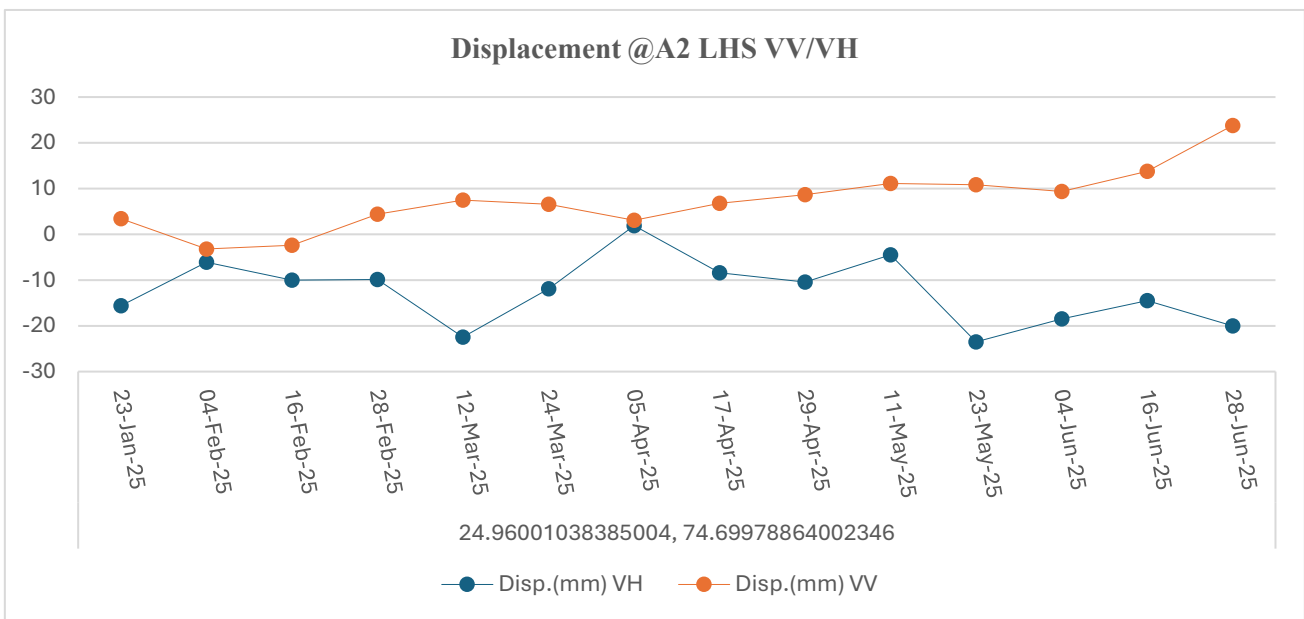
Critical Events & Anomalies:

- **January 2025:** VH starts near –10 mm, while VV is also negative (–12 mm), indicating early coupled lateral and vertical settlement.
- **May 2025:** VH recovers briefly to –1 mm, while VV remains strongly negative (–15 mm), indicating partial lateral recovery without vertical relief.
- **June 2025:** Both VH and VV remain negative (VH –10 mm, VV –16 mm), confirming persistent settlement behaviour toward the end of the monitoring period.

3.5 SAR Based Analysis at Abutment 2 LHS:



A2 LHS (24.96001038385004, 74.69978864002346)



Key Observations:

Displacement Magnitude Patterns:

- **Horizontal displacement (VH)** Horizontal displacement exhibits high variability with a strong negative dominance, ranging roughly between -5 mm and -25 mm.
- **Vertical Displacement (VV)** Vertical displacement shows a clear predominantly positive trend over the monitoring period, indicating overall vertical uplift at the A2 LHS location.

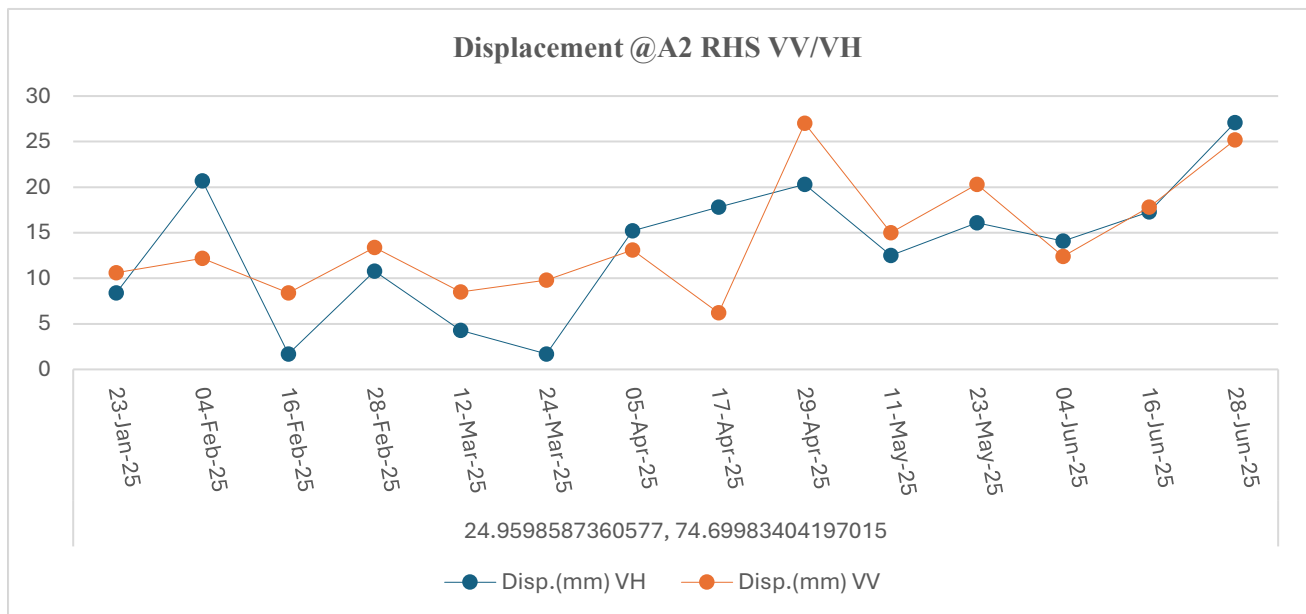
Critical Events & Anomalies:

- **April 2025:** VH drops again to -10 mm, while VV continues to rise ($+8$ mm), suggesting ongoing lateral instability despite improving vertical behaviour.
- **June 2025:** VH improves slightly to -18 to -15 mm, while VV increases sharply to $+14$ – 24 mm, confirming continued uplift intensification with unresolved lateral settlement toward the end of the monitoring period.

3.6 SAR Based Analysis at Abutment 2 RHS:



A2 RHS (24.9598587360577, 74.69983404197015)



Key Observations:

Displacement Magnitude Patterns:

- **Vertical (VV) displacement** Vertical displacement exhibits moderate to high variability with a predominantly positive trend over the monitoring period (Jan–Jun 2025), indicating overall vertical uplift at the A2 RHS location.
- **Horizontal (VH) displacement** Horizontal displacement shows high variability with a generally positive dominance, ranging roughly between +3 mm and +26 mm.

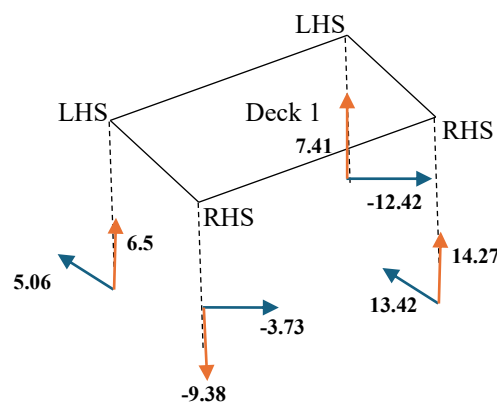
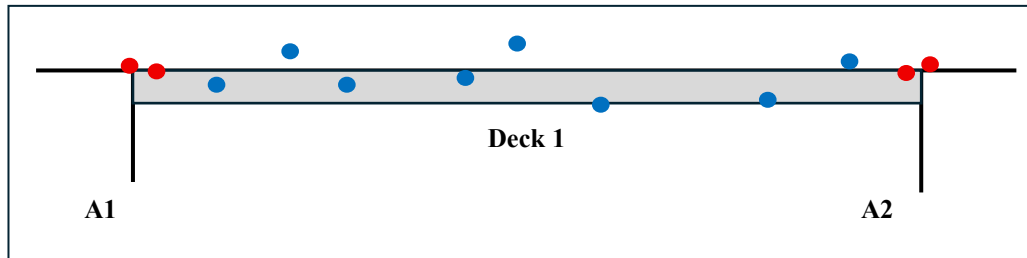
Critical Events & Anomalies:

- **April 2025:** VH increases to +15 mm, while VV drops to +6 mm, indicating a decoupled response with lateral dominance.
- **June 2025:** VH rises steadily to +26 mm, while VV increases to +25 mm, confirming continued coupled uplift and lateral displacement toward the end of the monitoring period.

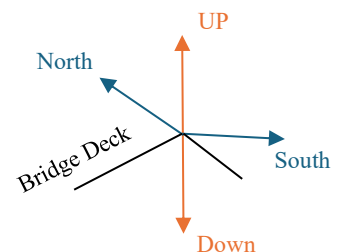
ANALYSIS

1. Identify potential deformation, shifts or movement of the bridge structure

- Points of Analysis (POA) of bridge (●)



Bridge Deck Slab Analysis



Vertical Displacement Behaviour:

- The bridge deck slab exhibits distinctly non-uniform vertical displacement, indicating differential vertical response between the LHS and RHS regions. The maximum vertical uplift is observed at the RHS support region, reaching approximately +14.27 mm, representing the dominant vertical response of the deck.

Horizontal Displacement Behaviour:

- Horizontal displacement demonstrates clear directional variability across the deck width, reflecting torsional deformation behaviour. The presence of opposing horizontal displacement directions across both width and depth confirms torsional rotation of the deck slab about a longitudinal axis.

CONCLUSION:

- The bridge deck slab exhibits a torsion-dominated deformation response, characterized by simultaneous vertical uplift and localized settlement combined with asymmetric horizontal displacement. Overall, the magnitude and asymmetry of the observed displacements suggest structurally significant behaviour, warranting continued monitoring and correlation with loading, bearing response, and environmental effects.